

Real Time Fraud Detection Smart Glasses

Capstone Project Proposal

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BE Third Year- COE/COPC

CPG No. 311

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MENTOR CONSENT FORM

I hereby agree to be the mentor of the following Capstone Project Team

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Mentor Consent form for capstone project of our team. (CPG no.: 311) [Inbox x](#)



JEEVANT VERMA

Respected Ma'am, I hope you are doing well. Please find attached the Mentor Consent Form for our capstone project titled "Realtime Fraud Detection Smart Glasses"



TARUNPREET BHATIA

to me ▾

Approved

Regards
Tarunpreet Bhatia
Associate Dean of Strategic Initiatives
Associate Professor, CSED
Thapar Institute of Engineering and Technology
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PROJECT OVERVIEW

The rapid advancement of artificial intelligence has led to a significant increase in sophisticated identity fraud. Technologies such as voice cloning, deepfake facial impersonation, and manipulated digital documents have made it easier for attackers to impersonate individuals. In recent years, nearly half of organizations have reported encountering deepfake-based fraud attempts, while digital document forgery has also seen a sharp global rise. These developments pose serious risks in areas such as financial transactions, identity verification, and digital communication.

Most existing verification systems rely on single-factor authentication methods such as passwords, OTPs, or basic facial recognition. However, these methods are increasingly vulnerable to advanced spoofing techniques, especially in sectors like banking and online services where identity verification is critical. Fraud is often detected only after the interaction has already taken place, leading to financial loss, identity theft, and misuse of personal information. Additionally, current systems typically evaluate voice, facial identity, and documents separately, creating gaps that attackers can exploit.

To address these challenges, this project proposes a wearable fraud detection solution integrated into smart glasses. The system is designed to support real-time identity verification during live interactions, enabling quicker and more reliable decision-making. By improving the effectiveness of verification processes in practical scenarios, the proposed solution aims to reduce the risk of modern AI-driven fraud and enhance trust in both digital and physical authentication systems.

PROBLEM STATEMENT

Identity verification systems are increasingly challenged by advanced AI-driven fraud techniques that can mimic human identity with high accuracy. These attacks exploit weaknesses in traditional authentication methods, enabling fraudsters to bypass security systems using synthetic voice, manipulated facial data, and forged digital documents. Additionally, many fraud attempts involve contextual inconsistencies, such as mismatched locations or unusual transaction patterns, which often go undetected.

Despite advancements in biometric technologies, existing solutions remain limited in effectiveness. Most systems operate on a single modality or evaluate different identity factors independently, resulting in fragmented verification processes. This lack of integration makes them vulnerable to sophisticated attacks that combine multiple forms of deception. Furthermore, many systems depend on delayed processing and centralized handling of sensitive data, reducing their ability to provide immediate and secure verification during real-world interactions.

The core problem addressed in this project is the absence of a unified and practical system capable of performing reliable identity verification in real time. There is a need for a solution that can simultaneously analyze multiple identity signals, detect inconsistencies, and provide immediate feedback while maintaining efficiency, privacy, and ease of use in real-world scenarios.

NEED ANALYSIS

The rapid advancement of artificial intelligence has significantly transformed the landscape of identity fraud, making it more sophisticated and harder to detect. Modern fraud techniques now include AI-driven voice cloning, deepfake facial manipulation, high-resolution ID replays, and synthetic identity mismatches. These developments have weakened traditional verification systems that rely on human inspection, static biometrics, or single-factor authentication. As digital identity becomes increasingly central to sectors like banking, border control, law enforcement, and corporate security, the scale and impact of identity fraud continue to grow.

Traditional verification methods suffer from several structural limitations. Manual ID checks are prone to human error and bias, while OTP and SMS-based authentication systems remain vulnerable to phishing and SIM swap attacks. Even standalone biometric systems, such as facial or voice recognition, can be bypassed using replay attacks or AI-generated inputs. Additionally, most systems function in isolation, verifying only a single data point without incorporating contextual information like user behavior or location, making them ineffective against multi-layered and coordinated fraud strategies.

Another key challenge is the lack of discreet, real-time verification in practical environments such as banking counters, immigration checkpoints, and field operations. Existing solutions often rely on bulky hardware or delayed processing, disrupting workflows and slowing decision-making. Furthermore, transmitting raw audio or video data to centralized servers raises privacy concerns under regulations like GDPR and India's DPDP Act. To address these issues, the proposed Wearable Real-Time Smart Fraud Detection Glasses System integrates multi-modal detection within a compact eyeglass platform, using a hybrid Edge-Gateway-Cloud architecture for efficient, privacy-preserving, and real-time fraud detection, along with subtle feedback through bone conduction audio.

LITERATURE SURVEY

This project proposes a multi-modal, wearable fraud detection system operating across three layers: an ESP32-S3 Sense edge chip in smart glasses, a smartphone gateway, and a cloud inference server. The system simultaneously detects AI-generated voice clones, deepfake facial impersonations, and fraudulent documents. This survey covers four core technical domains, identifies gaps, and explains how the system addresses each.

Voice Anti-Spoofing (Audio Liveness Module)

The ASVspoof challenge series is the global benchmark for anti-spoofing countermeasures (CMs) against TTS synthesis, voice conversion, and replay attacks. Todisco et al. [1] introduced ASVspoof 2019, the first edition addressing all three attack types and introducing the tandem detection cost function (t-DCF). Wang et al. [2] published the full ASVspoof 2019 corpus, now the standard CM benchmark, while ASVspoof 5 [3] expanded coverage to 32 attack algorithms with best-published Equal Error Rates around 6.3%. Published CMs range from CNN-based CQCC/MFCC models [4] to Transformer-based wav2vec and HuBERT architectures all assuming GPU or CPU deployment. No prior work explores microcontroller-class CM inference, the specific constraint of this system, which runs TensorFlow Lite Micro feature extraction on-chip and offloads full inference to the cloud.

Face Verification and Deepfake Detection (Face-to-Document Sync Module)

Schroff et al. [5] introduced FaceNet, producing compact 128-dimensional embeddings achieving 99.63% accuracy on LFW, suitable for mobile deployment. Deng et al. [6] proposed ArcFace with Additive Angular Margin Loss, reaching 99.40% on LFW and outperforming prior methods across ten benchmarks. On the forgery-detection front, Tolosana et al. [7] surveyed face manipulation categories synthesis, identity swap, attribute and expression manipulation and benchmarked detection approaches across each. Rössler et al. [8] introduced FaceForensics++,

comprising 1,000 videos manipulated with four state-of-the-art methods, demonstrating that domain-specific knowledge substantially improves detection under compression. A critical gap remains: matching a live face from a wearable camera to a low-resolution, aged ID photo is largely unstudied. This system addresses this using FaceNet embeddings at practical wearable capture distances.

Document Verification (Identity Cards and Official Documents)

Zanardelli et al. [9] surveyed deep learning approaches to image forgery detection covering copy-move, splicing, and inpainting, including CNN-based methods applied to document images. Boned et al. [10] introduced the SIDTD benchmark, a large-scale synthetic dataset of ID and travel documents with forgeries including portrait substitution and text-field replacement addressing the scarcity of real-document data due to privacy constraints. Tapia et al. [11] systematically reviewed over fifty identity card PAD methods across print, screen-replay, and digital-manipulation attack types, finding that all assume stationary, high-resolution cameras. This system fills that gap by assessing both ID cards and official documents under wearable off-angle capture conditions.

TinyML and Hybrid Edge-Cloud Architecture

TinyML enables ML inference on microcontrollers with kilobytes of RAM and milliwatt power budgets [12]. Capogrosso et al. [13] surveyed 779 TinyML studies, identifying privacy and latency as the primary motivations for on-device inference, with extraction latency typically under 100 ms. The Seeed Studio XIAO ESP32-S3 Sense 21×17.5 mm, 240 MHz dual-core, 8 MB PSRAM, built-in camera and microphone sits at the frontier of TinyML-capable biometric hardware. Velasquez-Henao et al. [14] identify the hybrid edge-cloud pattern as established for resource-constrained deployments. However, no published system integrates simultaneous audio-visual biometric feature extraction on a single microcontroller; this project is the first to do so in a wearable glasses form factor.

Multi-Modal KYC and Fraud Detection Systems

Regula Forensics [15] reports that 49% of organizations experienced deepfake attacks in 2024 (up from 29% in 2022), alongside a 244% rise in document forgeries, with over 70 billion identity verifications conducted globally. These trends have driven research toward integrated pipelines combining voice liveness, face matching, and document verification. However, all published multi-modal fraud detection systems are entirely server-side, with no edge component, no wearable form factor, and no covert result-delivery mechanism. This project is the first to combine edge TinyML extraction, smartphone gateway relay, cloud multi-modal inference, and bone conduction audio feedback within a standard glasses frame.

OBJECTIVES OF THE STUDY

1. To design a wearable fraud detection system integrated within smart glasses using an Edge-Gateway-Cloud framework. The system should ensure efficient communication and scalable processing for identity verification tasks.
2. To implement on-device feature extraction for audio and visual inputs to enable lightweight and efficient data processing. This approach aims to reduce dependency on raw data transmission while improving privacy and system responsiveness.
3. To develop a unified multi-modal verification system that combines voice, facial, and document-based authentication. The integration of multiple modalities enhances detection accuracy and reduces vulnerabilities of single-factor systems.
4. To enable real-time fraud detection during live interactions by generating immediate risk assessments. The system should support quick and reliable decision-making in practical identity verification scenarios as well as generating relevant analytics.

METHODOLOGY

Overview

The system follows a six-step pipeline spanning three architectural layers: the Edge Layer (glasses hardware), the Gateway Layer (smartphone bridge), and the Cloud Layer (AI inference). Each step builds upon the previous, ensuring real-time, multi-modal fraud detection with minimal user disruption.

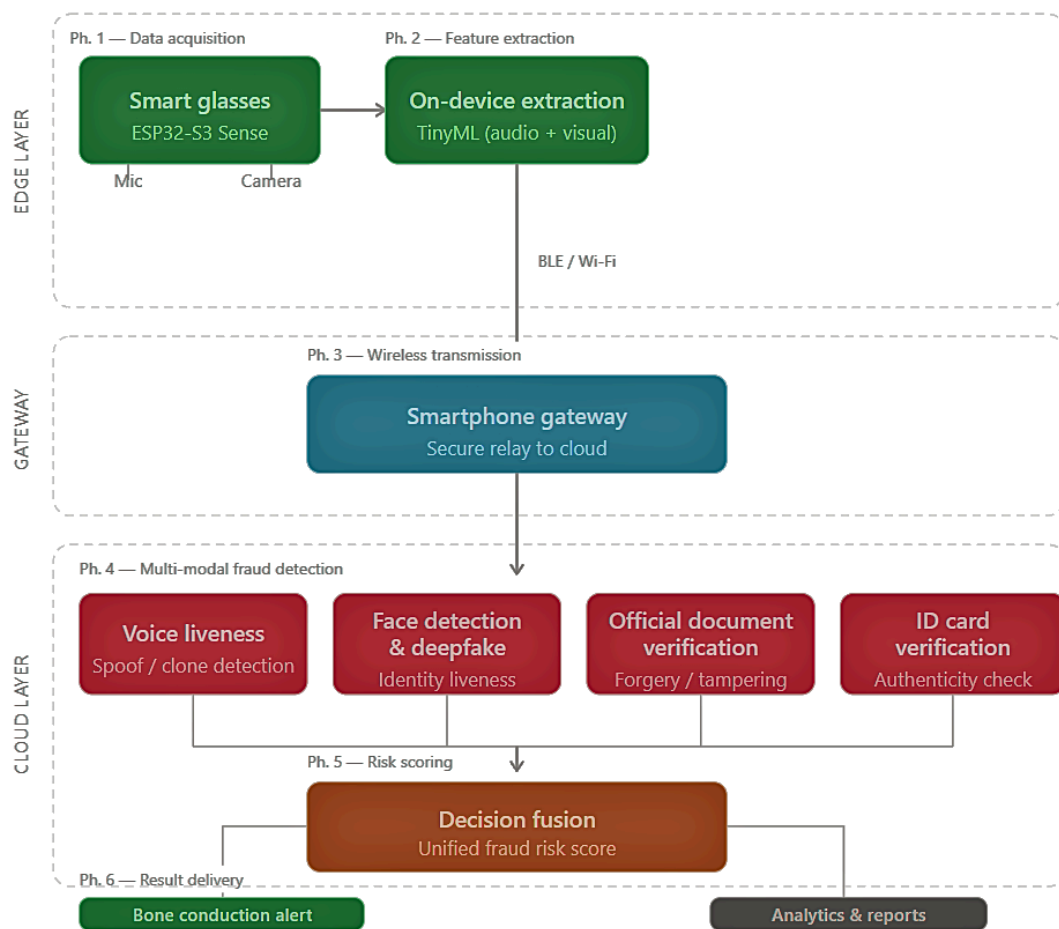


Fig. 1.1 Block Diagram of Methodology

Phase-wise Methodology:

Phase 1: Data Acquisition (Edge Layer)

The smart glasses, powered by the ESP32-S3 Sense microcontroller, continuously

capture audio and video using the built-in microphone and camera. Data is collected in real time during identity verification interactions.

Phase 2: On-Device Feature Extraction (TinyML)

Lightweight models process the captured inputs directly on the device to extract relevant features from audio and image data. Only compressed feature representations are generated, reducing the need to transmit raw data and improving efficiency and privacy.

Phase 3: Wireless Transmission to Gateway

The extracted feature data is transmitted to a paired smartphone via wireless communication such as BLE or Wi-Fi. The smartphone acts as a gateway and securely forwards the data to the cloud server for further processing.

Phase 4: Multi-Modal Fraud Detection (Cloud Layer)

The cloud system performs detailed analysis using multiple verification modules, including voice-based liveness detection, document authenticity verification, facial identity matching, and contextual validation.

Phase 5: Decision Fusion and Risk Scoring

Outputs from all verification modules are combined to generate a unified fraud risk score. Based on this score, the system classifies the interaction as legitimate, suspicious, or fraudulent.

Phase 6: Result Delivery via Bone Conduction and Analytics Generation

The final result is converted into a brief audio alert and transmitted back to the smart glasses. Using bone conduction, the system provides discreet feedback to the user without interrupting the interaction. Detailed analysis and reports are made available on the connected smartphone.

PROJECT OUTCOMES & INDIVIDUAL ROLES

Project Outcomes / Deliverables:

The primary outcome of this capstone project will be the development of a functional prototype of a wearable real-time fraud detection system integrated within a glasses-based form factor. The system will demonstrate the feasibility of using the Sreed Studio XIAO ESP32-S3 Sense to capture audio and visual inputs while performing lightweight edge-based feature extraction using TinyML techniques. These extracted features will be transmitted through a smartphone gateway to a cloud-based backend where more computationally intensive verification algorithms will analyze the data.

The software component of the project will implement a multi-modal fraud detection pipeline that integrates voice liveness detection, document authenticity verification, face-to-ID matching, and contextual anomaly analysis to produce a unified fraud confidence assessment. The system will operate using a hybrid Edge–Gateway–Cloud architecture, enabling real-time decision-making while minimizing raw biometric data transmission. To ensure discreet operation during real-world interactions, the final verification result will be communicated to the user through bone conduction audio alerts, allowing silent and non-intrusive feedback.

In addition to the working prototype, the project will also produce supporting software infrastructure, system architecture documentation, performance evaluation results, and presentation materials that demonstrate the practicality and effectiveness of the proposed wearable fraud detection framework.

Individual Roles and Responsibilities:

The project is implemented as a collaborative effort in which team members contribute to system design, software development, and documentation. Responsibilities are distributed according to technical expertise in full-stack development, artificial intelligence, and system architecture.

Table 1 - Roles and Responsibilities

Sr. No.	Feature	Assigned To
1.	Smart glasses (ESP32-S3 Sense) & On-device extraction (TinyML audio + visual)	Yuvansh Pathak, Jeevant Verma
2.	Smartphone gateway (Secure relay to cloud) & App development	Aniket Saxena, Yuvansh Pathak
3.	Voice liveness (Anti-spoofing) & Face matching (Face-to-ID sync)	Jeevant Verma, Lakshita Gupta
4.	Context check (Location / anomaly) & Decision fusion (Unified fraud risk score)	Aniket Saxena, Mannat Kaur Miglani
5.	Analytics & reports	Lakshita Gupta, Aniket Saxena

WORK PLAN

The following Gantt chart outlines the planned timeline and actual progress for achieving the project objectives across an 18-week period (July – December).

Table 2 - Work Plan Gantt chart

Sr. No.	Activity	Month	July		August				September				November				December			
		Week	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18
1	Identification, Formulation and Planning of Project	Plan																		
2	Literature Review and Technology Analysis	Plan																		
3	System Architecture Design & Communication Study	Plan																		
4	Data Collection and Preprocessing	Plan																		
5	Model Development and Validation	Plan																		
6	ESP32-S3 TinyML Model Testing and Calibration	Plan																		
7	Hardware Interfacing (Glasses Assembly)	Plan																		
8	System Integration and Communication Implementation	Plan																		
9	Design Optimization (Power and Latency)	Plan																		
10	Integration Testing	Plan																		
11	Results Evaluation and Validation	Plan																		
12	Final Report Preparation and Documentation	Plan																		

COURSE SUBJECT MAPPING FOR THE PROPOSED SYSTEM

Table 3 - Subject Mapping

S. No.	Course Subject	Concepts Applied	Role in Proposed System
1	Artificial Intelligence	Classification models, decision-making systems, risk scoring	Enables fraud confidence computation and multi-modal decision fusion
2	Machine Learning	Supervised & unsupervised learning, anomaly detection, model evaluation metrics (FAR, FRR, EER)	Used for voice spoof detection, contextual anomaly detection, and performance benchmarking
3	Deep Learning	CNN, LSTM, embedding models, neural optimization	Applied in audio liveness detection, document verification, and face embedding extraction
4	Computer Networks	Secure API communication, latency management, encrypted transmission	Ensures reliable and secure transfer of feature vectors to cloud infrastructure
5	Cloud Computing	Distributed computing, backend deployment (FastAPI), scalable inference	Performs heavy deep learning inference and centralized decision fusion
6	Cyber Security	Spoof attack modeling, adversarial threats, secure data handling	Strengthens system robustness against AI-driven fraud and ensures privacy compliance

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